

paper

Arezoo

0. Setup

```
library(AER)
library(BayesFactor)
library(BFpack)
library(brms)
library(broom)
library(corrplot)
library(dagitty)
library(devtools)
library(dplyr)
library(easystats)
library(emmeans)
library(GGally)
library(ggdag)
library(ggdist)
library(ggExtra)
library(ggplot2)
library(ggrepel)
library(GPARotation)
library(gridExtra)
library(gtsummary)
library(haven)
library(infer)
library(janitor)
library(jpeg)
library(kableExtra)
library(knitr)
library(lavaan)
library(lme4)
```

```
library(logspline)
library(magrittr)
library(margins)
library(modelr)
library(patchwork)
library(pbkrtest)
library(psych)
library(purrr)
library(pwr)
library(readr)
library(readxl)
library(rstanarm)
library(scales)
library(semPlot)
library(stringr)
library(tidyLPA)
library(tidyr)
library(tidyverse)
```

Data

```
# adding data set
data <- read.csv("data.csv")

data$sex <- as.factor(data$sex)
data$ethnicity <- as.factor(data$ethnicity)
data$year <- as.factor(data$year)

#rename satisfactions
data <- dplyr::rename(
  data,
  `Security` = AR_1_1,
  `Acceptance` = AR_1_2,
  `Initiative` = AR_1_3,
  `Competence` = AR_1_4,
  `Meaning` = AR_1_5)
```

LPA

Models

```
data%>%
  dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles=1:2,
                   variances = "varying",
                   covariances = "varying") ->model6_all
```

Warning in estimate_profiles.default(., n_profiles = 1:2, variances = "varying", : Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values, which may violate this assumption. Carefully examine the results, and avoid complex models, e.g.with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

Warning in (function (data, modelName = NULL, nboot = 999, level = 0.05, : some model(s) could not be fitted!

```
model6_all
```

tidyLPA analysis using mclust:

	Model	Classes	AIC	BIC	Entropy	prob_min	prob_max	n_min	n_max	BLRT_p
1	6	1	6406.74	6525.73	1.00	1.00	1.00	1.00	1.00	
2	6	2	6155.84	6397.78	0.69	0.90	0.92	0.46	0.54	0.01

```
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles=1:3,
                   variances = "varying",
                   covariances = "zero") ->model2_all
```

Warning in estimate_profiles.default(., n_profiles = 1:3, variances = "varying", : Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values,

which may violate this assumption. Carefully examine the results, and avoid complex models, e.g. with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

Warning in (function (data, modelName = NULL, nboot = 999, level = 0.05, : some model(s) could not be fitted!

```
model2_all
```

tidyLPA analysis using mclust:

	Model	Classes	AIC	BIC	Entropy	prob_min	prob_max	n_min	n_max	BLRT_p
1	2	1	6792.67	6832.33	1.00	1.00	1.00	1.00	1.00	
2	2	2	6338.22	6421.51	0.75	0.89	0.96	0.43	0.57	0.01
3	2	3	6230.51	6357.43	0.76	0.87	0.91	0.16	0.52	0.01

```
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles=1:6,
                   variances = "equal",
                   covariances = "zero") ->model1_all
```

Warning in estimate_profiles.default(., n_profiles = 1:6, variances = "equal", : Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values, which may violate this assumption. Carefully examine the results, and avoid complex models, e.g. with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

```
model1_all
```

tidyLPA analysis using mclust:

	Model	Classes	AIC	BIC	Entropy	prob_min	prob_max	n_min	n_max	BLRT_p
1	1	1	6792.67	6832.33	1.00	1.00	1.00	1.00	1.00	
2	1	2	6464.16	6527.62	0.79	0.88	0.96	0.29	0.71	0.01
3	1	3	6388.84	6476.09	0.73	0.84	0.92	0.13	0.46	0.01
4	1	4	6396.42	6507.48	0.70	0.75	0.91	0.20	0.31	0.70
5	1	5	6352.04	6486.89	0.73	0.65	0.91	0.08	0.37	0.01
6	1	6	6304.93	6463.58	0.74	0.77	0.94	0.08	0.31	0.01

Selected Models

```
#Model2_Class 2 & 3
#Model 1_class 2 & 3 & 6

#####
##Model2_Class 2
#####
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles = 2,
                   variances = "varying",
                   covariances = "zero") ->model2_2
```

Warning in estimate_profiles.default(., n_profiles = 2, variances = "varying",
: Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values, which may violate this assumption. Carefully examine the results, and avoid complex models, e.g. with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

```
#Parameter estimates
print(tidyLPA::get_estimates(model2_2), n=28)
```

A tibble: 20 x 8

	Category	Parameter	Estimate	se	p	Class	Model	Classes
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>
1	Means	Security	4.71	0.141	3.44e-244	1	2	2
2	Means	Acceptance	3.75	0.151	8.13e-136	1	2	2
3	Means	Initiative	3.86	0.158	1.91e-132	1	2	2
4	Means	Competence	4.75	0.117	0	1	2	2
5	Means	Meaning	4.27	0.147	2.91e-186	1	2	2
6	Variances	Security	1.97	0.195	5.29e- 24	1	2	2
7	Variances	Acceptance	1.90	0.201	3.75e- 21	1	2	2
8	Variances	Initiative	1.89	0.203	1.20e- 20	1	2	2
9	Variances	Competence	1.78	0.215	1.30e- 16	1	2	2
10	Variances	Meaning	2.21	0.216	1.37e- 24	1	2	2
11	Means	Security	6.07	0.0987	0	2	2	2
12	Means	Acceptance	5.08	0.127	0	2	2	2
13	Means	Initiative	5.72	0.118	0	2	2	2

14 Means	Competence	5.95	0.0854	0		2	2	2
15 Means	Meaning	5.73	0.120	0		2	2	2
16 Variances	Security	0.644	0.122	1.17e- 7		2	2	2
17 Variances	Acceptance	1.93	0.202	1.27e- 21		2	2	2
18 Variances	Initiative	0.669	0.107	4.22e- 10		2	2	2
19 Variances	Competence	0.486	0.0600	5.53e- 16		2	2	2
20 Variances	Meaning	0.849	0.150	1.55e- 8		2	2	2

```
round(model2_2$model_2_class_2$fit, digits=3)
```

	Model	Classes	LogLik	parameters	n	AIC	AWE	BIC	CAIC
1	2	2	-3148.112		21 390	6338.225	6608.312	6421.514	6442.514
		CLC	KIC	SABIC	ICL	Entropy	prob_min	prob_max	n_min n_max
1	6297.715	6362.225	6354.882	-6487.614	0.745	0.894	0.956	0.431	0.569
	BLRT_val	BLRT_p							
1	476.446	0.01							

```
#Or
get_fit(model2_2)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters      n    AIC    AWE    BIC    CAIC    CLC    KIC
  <dbl>    <dbl>  <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1      2      2 -3148.      21    390 6338. 6608. 6422. 6443. 6298. 6362.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

```
#####
##Model2_Class 3
#####
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles = 3,
                    variances = "varying",
                    covariances = "zero") ->model2_3
```

Warning in estimate_profiles.default(., n_profiles = 3, variances = "varying",
: Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values, which may violate this assumption. Carefully examine the results, and avoid complex models, e.g. with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

Warning in (function (data, modelName = NULL, nboot = 999, level = 0.05, : some model(s) could not be fitted!

```
#Parameter estimates
print(tidyLPA::get_estimates(model2_3), n=28)
```

A tibble: 30 x 8

	Category	Parameter	Estimate	se	p	Class	Model	Classes
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>
1	Means	Security	4.57	0.156	3.56e-188	1	2	3
2	Means	Acceptance	3.63	0.243	1.81e- 50	1	2	3
3	Means	Initiative	3.64	0.272	6.75e- 41	1	2	3
4	Means	Competence	4.56	0.190	5.35e-127	1	2	3
5	Means	Meaning	4.09	0.230	6.72e- 71	1	2	3
6	Variances	Security	2.17	0.233	1.03e- 20	1	2	3
7	Variances	Acceptance	1.93	0.342	1.86e- 8	1	2	3
8	Variances	Initiative	2.05	0.329	4.36e- 10	1	2	3
9	Variances	Competence	2.04	0.282	4.61e- 13	1	2	3
10	Variances	Meaning	2.49	0.321	7.94e- 15	1	2	3
11	Means	Security	5.73	0.166	4.20e-261	2	2	3
12	Means	Acceptance	4.57	0.155	3.55e-190	2	2	3
13	Means	Initiative	5.30	0.147	1.21e-285	2	2	3
14	Means	Competence	5.69	0.104	0	2	2	3
15	Means	Meaning	5.32	0.121	0	2	2	3
16	Variances	Security	0.815	0.290	4.95e- 3	2	2	3
17	Variances	Acceptance	1.89	0.199	1.75e- 21	2	2	3
18	Variances	Initiative	0.769	0.198	1.00e- 4	2	2	3
19	Variances	Competence	0.455	0.113	6.12e- 5	2	2	3
20	Variances	Meaning	0.863	0.182	2.21e- 6	2	2	3
21	Means	Security	6.57	0.116	0	3	2	3
22	Means	Acceptance	6.10	0.415	6.04e- 49	3	2	3
23	Means	Initiative	6.28	0.141	0	3	2	3
24	Means	Competence	6.42	0.127	0	3	2	3
25	Means	Meaning	6.45	0.141	0	3	2	3
26	Variances	Security	0.321	0.0782	4.14e- 5	3	2	3
27	Variances	Acceptance	0.569	0.873	5.14e- 1	3	2	3
28	Variances	Initiative	0.353	0.0658	8.02e- 8	3	2	3

i 2 more rows

```
round(model2_3$model_2_class_3$fit, digits=3)
```

Model	Classes	LogLik	parameters	n	AIC	AWE	BIC	CAIC
-------	---------	--------	------------	---	-----	-----	-----	------

```

1      2      3 -3083.255      32 390 6230.51 6642.82 6357.427 6389.427
      CLC      KIC      SABIC      ICL Entropy prob_min prob_max n_min n_max
1 6168.034 6265.51 6255.893 -6454.358 0.762 0.868 0.91 0.159 0.515
      BLRT_val BLRT_p
1 129.714 0.01

```

```
#Or
```

```
get_fit(model2_3)
```

```
# A tibble: 1 x 20
```

```

  Model Classes LogLik parameters      n  AIC  AWE  BIC  CAIC  CLC  KIC
<dbl> <dbl> <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1      2      3 -3083.      32   390 6231. 6643. 6357. 6389. 6168. 6266.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>

```

```
#####
```

```
#Model 1_class 2
```

```
#####
```

```

data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles = 2,
                    variances = "equal",
                    covariances = "zero") ->model1_2

```

Warning in estimate_profiles.default(., n_profiles = 2, variances = "equal", : Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values, which may violate this assumption. Carefully examine the results, and avoid complex models, e.g. with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

```
#Parameter estimates
```

```
print(tidyLPA::get_estimates(model1_2), n=28)
```

```
# A tibble: 20 x 8
```

```

  Category Parameter Estimate      se      p Class Model Classes
<chr>      <chr>      <dbl> <dbl>    <dbl> <int> <dbl> <dbl>
1 Means    Security     4.66 0.142 1.68e-236 1      1      2
2 Means    Acceptance     3.48 0.121 3.27e-181 1      1      2

```


3	Means	Initiative	3.30	0.173	8.72e- 81	1	1	2
4	Means	Competence	4.47	0.177	7.02e-141	1	1	2
5	Means	Meaning	3.88	0.166	2.66e-121	1	1	2
6	Variances	Security	1.43	0.139	6.85e- 25	1	1	2
7	Variances	Acceptance	1.92	0.121	1.38e- 56	1	1	2
8	Variances	Initiative	0.982	0.0899	8.43e- 28	1	1	2
9	Variances	Competence	1.05	0.103	4.19e- 24	1	1	2
10	Variances	Meaning	1.38	0.101	1.26e- 42	1	1	2
11	Means	Security	5.80	0.0684	0	2	1	2
12	Means	Acceptance	4.91	0.104	0	2	1	2
13	Means	Initiative	5.57	0.0801	0	2	1	2
14	Means	Competence	5.82	0.0645	0	2	1	2
15	Means	Meaning	5.59	0.0736	0	2	1	2
16	Variances	Security	1.43	0.139	6.85e- 25	2	1	2
17	Variances	Acceptance	1.92	0.121	1.38e- 56	2	1	2
18	Variances	Initiative	0.982	0.0899	8.43e- 28	2	1	2
19	Variances	Competence	1.05	0.103	4.19e- 24	2	1	2
20	Variances	Meaning	1.38	0.101	1.26e- 42	2	1	2

```
round(model1_2$model_1_class_2$fit, digits=3)
```

	Model	Classes	LogLik	parameters	n	AIC	AWE	BIC	CAIC
1	1	2	-3216.081		16 390	6464.163	6669.498	6527.621	6543.621
		CLC	KIC	SABIC	ICL	Entropy	prob_min	prob_max	n_min n_max
1	6433.744	6483.163	6476.854	-6584.022	0.791	0.883	0.964	0.295	0.705
		BLRT_val	BLRT_p						
1	340.508	0.01							

```
#Or
get_fit(model1_2)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters      n  AIC  AWE  BIC  CAIC  CLC  KIC
  <dbl>   <dbl>   <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1     2 -3216.         16   390 6464. 6669. 6528. 6544. 6434. 6483.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
# prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

```
#####
#Model 1_class 3
#####
```

```
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles = 3,
                   variances = "equal",
                   covariances = "zero") ->model1_3
```

Warning in estimate_profiles.default(., n_profiles = 3, variances = "equal", : Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values, which may violate this assumption. Carefully examine the results, and avoid complex models, e.g. with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

```
#Parameter estimates
print(tidyLPA::get_estimates(model1_3), n=28)
```

A tibble: 30 x 8

	Category	Parameter	Estimate	se	p	Class	Model	Classes
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>
1	Means	Security	4.79	0.325	4.72e- 49	1	1	3
2	Means	Acceptance	3.35	0.308	1.82e- 27	1	1	3
3	Means	Initiative	2.40	0.245	1.32e- 22	1	1	3
4	Means	Competence	4.02	0.415	3.23e- 22	1	1	3
5	Means	Meaning	2.89	0.509	1.46e- 8	1	1	3
6	Variances	Security	1.37	0.163	3.67e- 17	1	1	3
7	Variances	Acceptance	1.83	0.202	1.60e- 19	1	1	3
8	Variances	Initiative	0.756	0.148	3.13e- 7	1	1	3
9	Variances	Competence	0.969	0.137	1.74e- 12	1	1	3
10	Variances	Meaning	1.10	0.179	8.56e- 10	1	1	3
11	Means	Security	4.98	0.255	5.91e- 85	2	1	3
12	Means	Acceptance	4.00	0.200	1.07e- 88	2	1	3
13	Means	Initiative	4.50	0.228	6.57e- 87	2	1	3
14	Means	Competence	5.13	0.167	3.60e-208	2	1	3
15	Means	Meaning	4.87	0.123	0	2	1	3
16	Variances	Security	1.37	0.163	3.67e- 17	2	1	3
17	Variances	Acceptance	1.83	0.202	1.60e- 19	2	1	3
18	Variances	Initiative	0.756	0.148	3.13e- 7	2	1	3
19	Variances	Competence	0.969	0.137	1.74e- 12	2	1	3
20	Variances	Meaning	1.10	0.179	8.56e- 10	2	1	3
21	Means	Security	6.09	0.123	0	3	1	3
22	Means	Acceptance	5.25	0.206	8.51e-143	3	1	3

```

23 Means      Initiative      5.92  0.126  0                3      1      3
24 Means      Competence     6.06  0.0919 0                3      1      3
25 Means      Meaning        5.86  0.150  0                3      1      3
26 Variances  Security        1.37  0.163  3.67e- 17        3      1      3
27 Variances  Acceptance        1.83  0.202  1.60e- 19        3      1      3
28 Variances  Initiative         0.756 0.148  3.13e- 7         3      1      3
# i 2 more rows

```

```
round(model1_3$model_1_class_3$fit, digits=3)
```

```

Model Classes      LogLik parameters      n      AIC      AWE      BIC      CAIC
1      1      3 -3172.418      22 390 6388.837 6671.888 6476.092 6498.092
      CLC      KIC      SABIC      ICL Entropy prob_min prob_max n_min n_max
1 6346.295 6413.837 6406.287 -6592.063 0.729 0.841 0.916 0.126 0.459
      BLRT_val BLRT_p
1 87.326 0.01

```

```

#Or
get_fit(model1_3)

```

```

# A tibble: 1 x 20
Model Classes LogLik parameters      n      AIC      AWE      BIC      CAIC      CLC      KIC
<dbl> <dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1      1      3 -3172.      22 390 6389. 6672. 6476. 6498. 6346. 6414.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
# prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>

```

```

#####
#Model 1_class 6
#####
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles = 6,
                    variances = "equal",
                    covariances = "zero") ->model1_6

```

Warning in estimate_profiles.default(., n_profiles = 6, variances = "equal", : Latent profile analysis assumes that all variables have a continuous measurement level. Some variables have fewer than 10 unique values, which may violate this assumption. Carefully examine the results, and avoid complex models, e.g. with free (co)variances or many classes. The variables violating this assumption are: Security, Acceptance, Initiative, Competence, Meaning

```
#Parameter estimates
print(tidyLPA::get_estimates(model1_6), n=28)
```

```
# A tibble: 60 x 8
```

	Category	Parameter	Estimate	se	p	Class	Model	Classes
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>
1	Means	Security	4.80	0.487	5.78e- 23	1	1	6
2	Means	Acceptance	3.30	0.436	3.80e- 14	1	1	6
3	Means	Initiative	2.21	0.209	3.22e- 26	1	1	6
4	Means	Competence	4.28	0.626	8.29e- 12	1	1	6
5	Means	Meaning	2.00	0.317	2.67e- 10	1	1	6
6	Variances	Security	1.13	0.235	1.48e- 6	1	1	6
7	Variances	Acceptance	1.11	0.344	1.21e- 3	1	1	6
8	Variances	Initiative	0.520	0.112	3.40e- 6	1	1	6
9	Variances	Competence	1.01	0.144	2.26e- 12	1	1	6
10	Variances	Meaning	0.808	0.126	1.36e- 10	1	1	6
11	Means	Security	5.50	0.805	8.45e- 12	2	1	6
12	Means	Acceptance	3.05	0.392	6.40e- 15	2	1	6
13	Means	Initiative	5.82	0.380	7.01e- 53	2	1	6
14	Means	Competence	5.73	0.396	2.40e- 47	2	1	6
15	Means	Meaning	5.81	0.353	9.53e- 61	2	1	6
16	Variances	Security	1.13	0.235	1.48e- 6	2	1	6
17	Variances	Acceptance	1.11	0.344	1.21e- 3	2	1	6
18	Variances	Initiative	0.520	0.112	3.40e- 6	2	1	6
19	Variances	Competence	1.01	0.144	2.26e- 12	2	1	6
20	Variances	Meaning	0.808	0.126	1.36e- 10	2	1	6
21	Means	Security	5.55	0.187	5.30e-193	3	1	6
22	Means	Acceptance	5.06	0.349	1.22e- 47	3	1	6
23	Means	Initiative	4.93	0.196	1.09e-139	3	1	6
24	Means	Competence	5.46	0.223	5.66e-133	3	1	6
25	Means	Meaning	4.86	0.380	2.23e- 37	3	1	6
26	Variances	Security	1.13	0.235	1.48e- 6	3	1	6
27	Variances	Acceptance	1.11	0.344	1.21e- 3	3	1	6
28	Variances	Initiative	0.520	0.112	3.40e- 6	3	1	6

```
# i 32 more rows
```

```
round(model1_6$model_1_class_6$fit, digits=3)
```

	Model	Classes	LogLik	parameters	n	AIC	AWE	BIC	CAIC	
1	1	6	-3112.465		40	390	6304.93	6820.738	6463.576	6503.576
	CLC	KIC	SABIC	ICL	Entropy	prob_min	prob_max	n_min	n_max	

```
1 6226.413 6347.93 6336.658 -6639.02 0.742 0.771 0.94 0.079 0.313
  BLRT_val BLRT_p
1 59.114 0.01
```

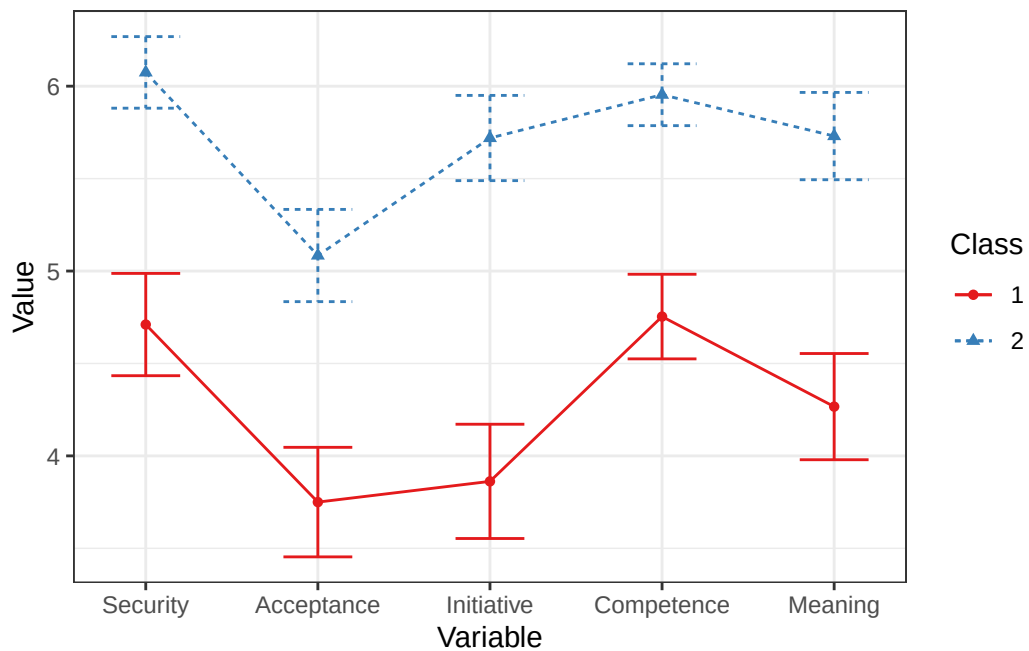
```
#Or
get_fit(model1_6)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters      n  AIC  AWE  BIC  CAIC  CLC  KIC
  <dbl>   <dbl>   <dbl>   <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1       6 -3112.       40   390 6305. 6821. 6464. 6504. 6226. 6348.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

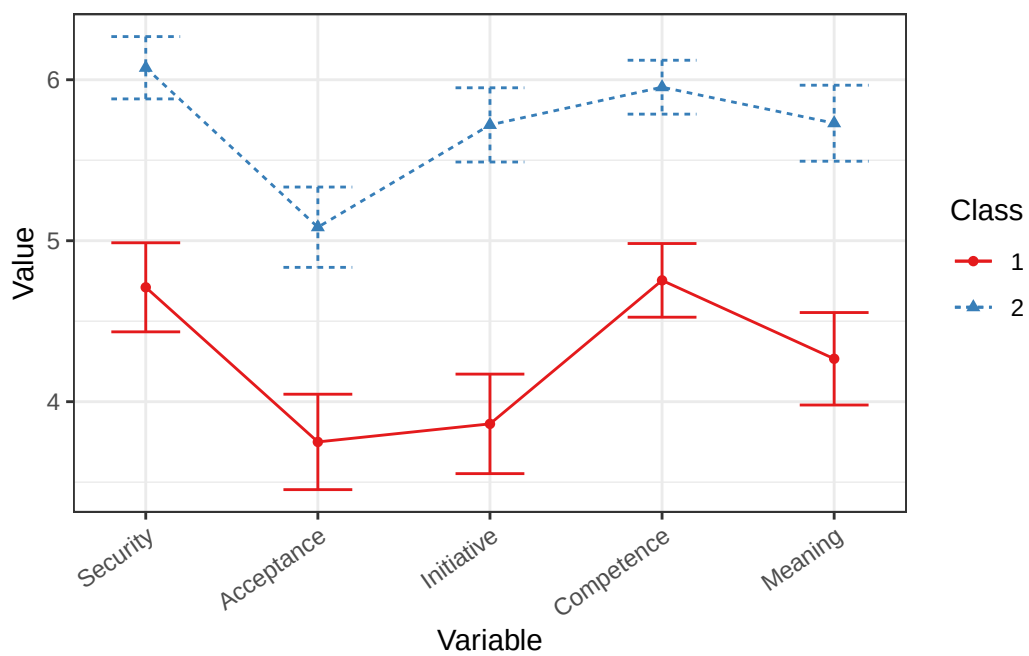
Plot

```
#####
# Model 2. Plot the two-class solution.
#####

tidyLPA::plot_profiles(
  model2_2,
  rawdata = FALSE, # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE      # Hide standard-deviation bars.
) +
  ggplot2::theme(
    # Rotate and align indicator labels to improve readability.
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```

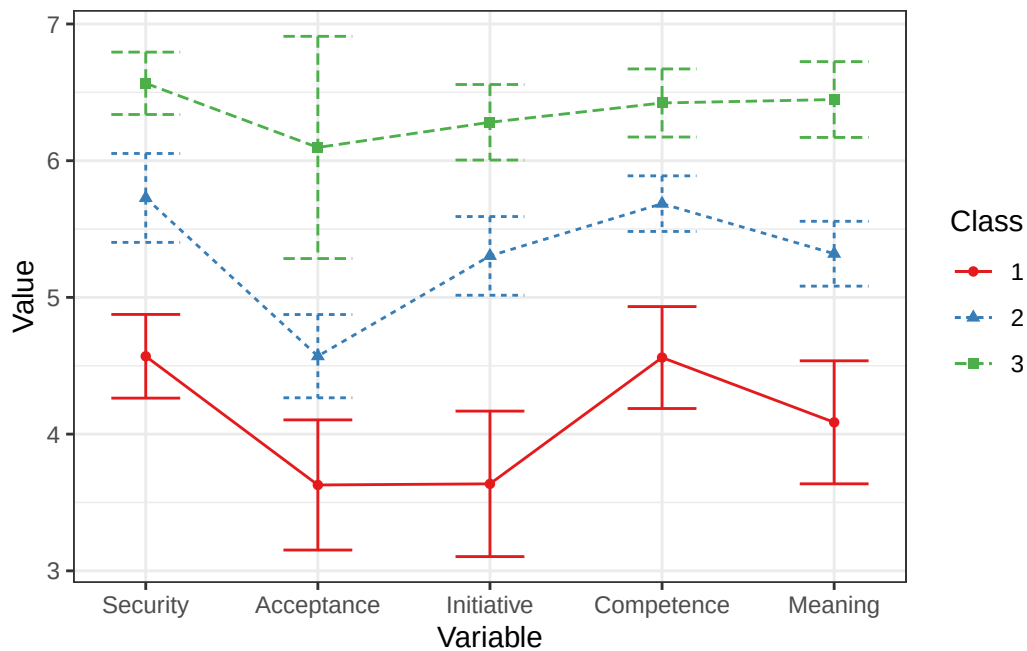


Warning: Using shapes for an ordinal variable is not advised

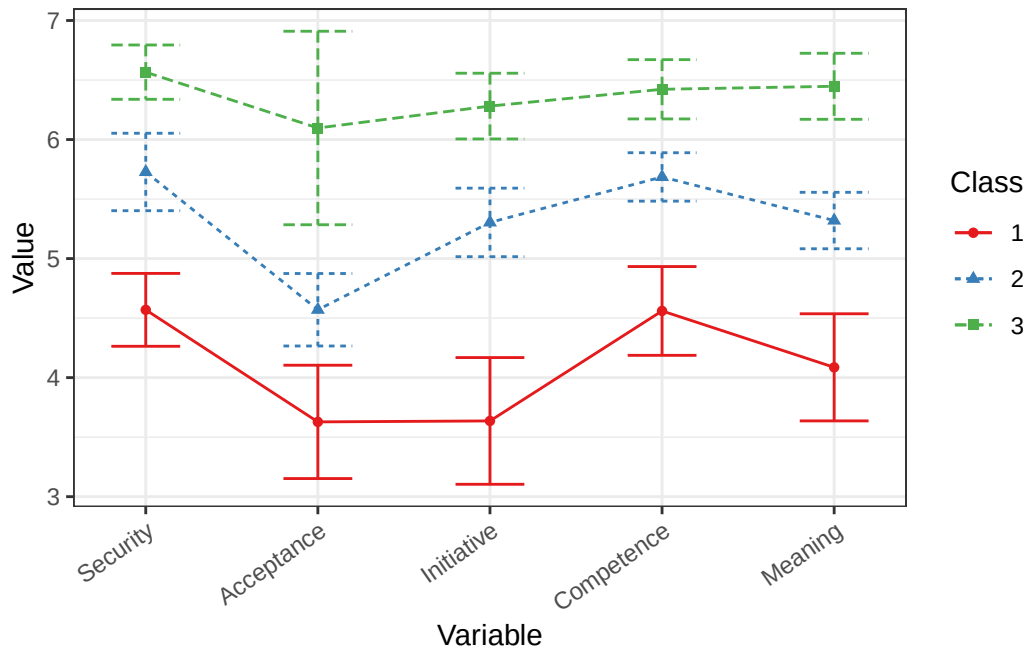


```
# -----
# For the other plots, replace "model_2_class_3" above with:
# -----

tidyLPA::plot_profiles(
  model2_3,
  rawdata = FALSE, # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE       # Hide standard-deviation bars.
) +
  ggplot2::theme(
    # Rotate and align indicator labels to improve readability.
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```

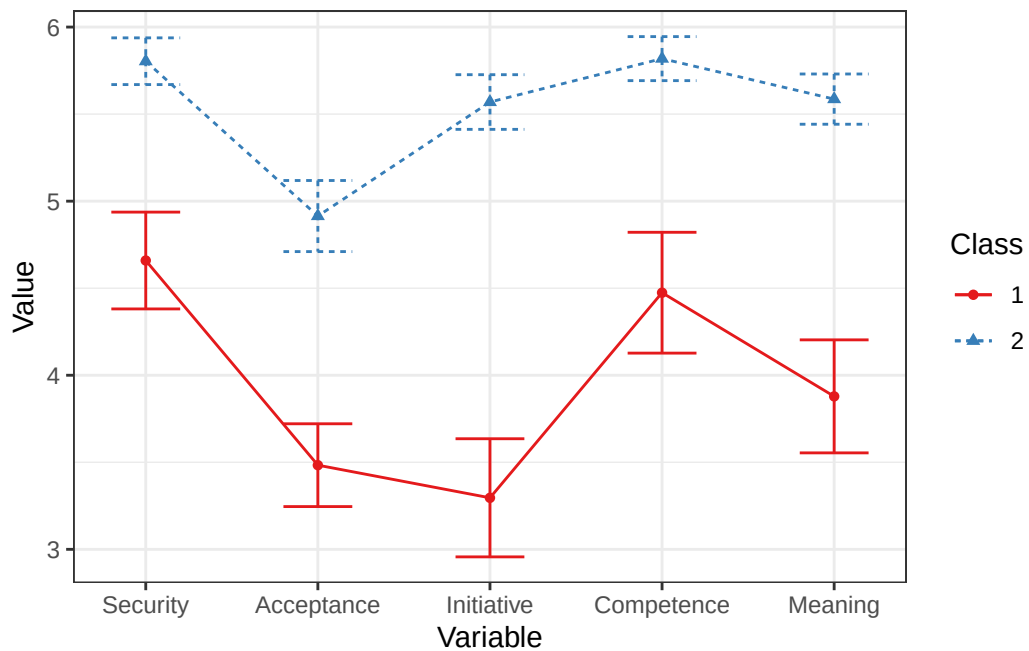


Warning: Using shapes for an ordinal variable is not advised

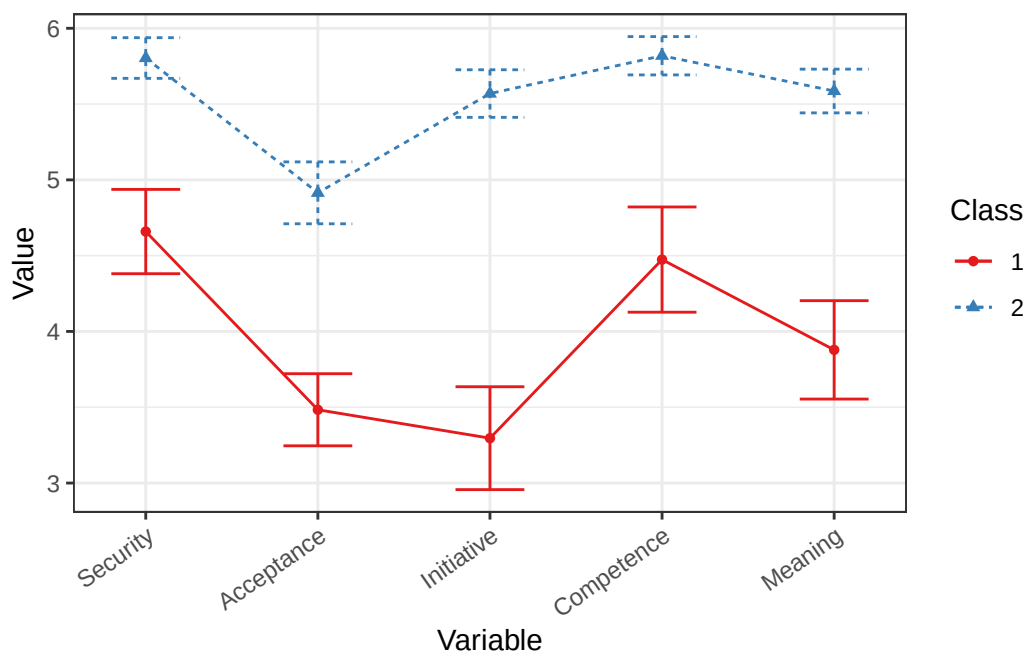


```
#####
#Model 1 Class 2 - 3 - 6
#####

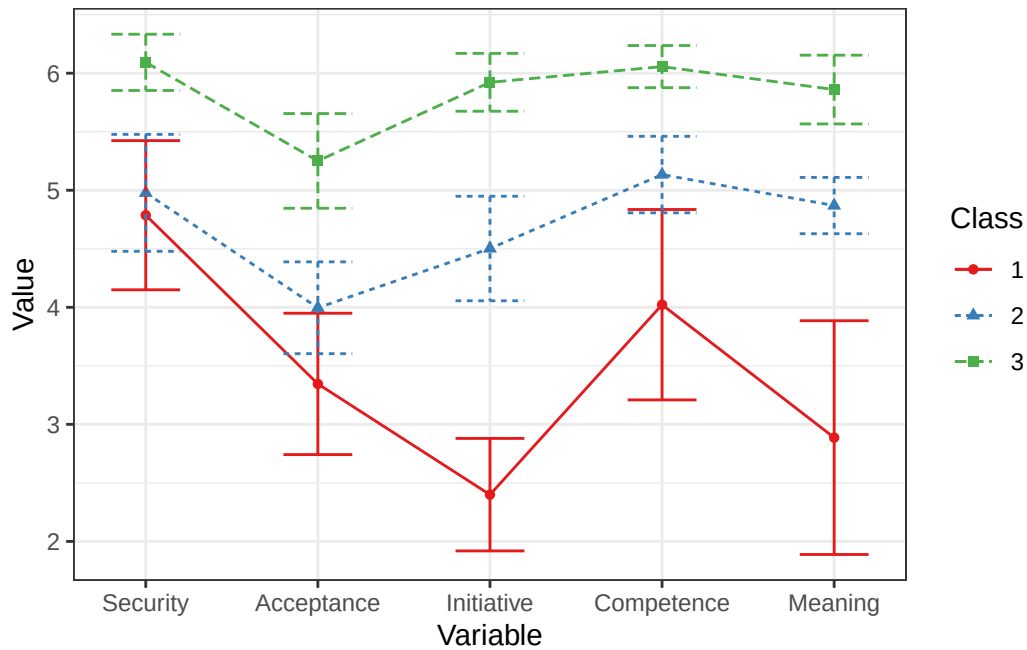
tidyLPA::plot_profiles(
  model1_2,
  rawdata = FALSE, # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE      # Hide standard-deviation bars.
) +
ggplot2::theme(
  axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
)
```

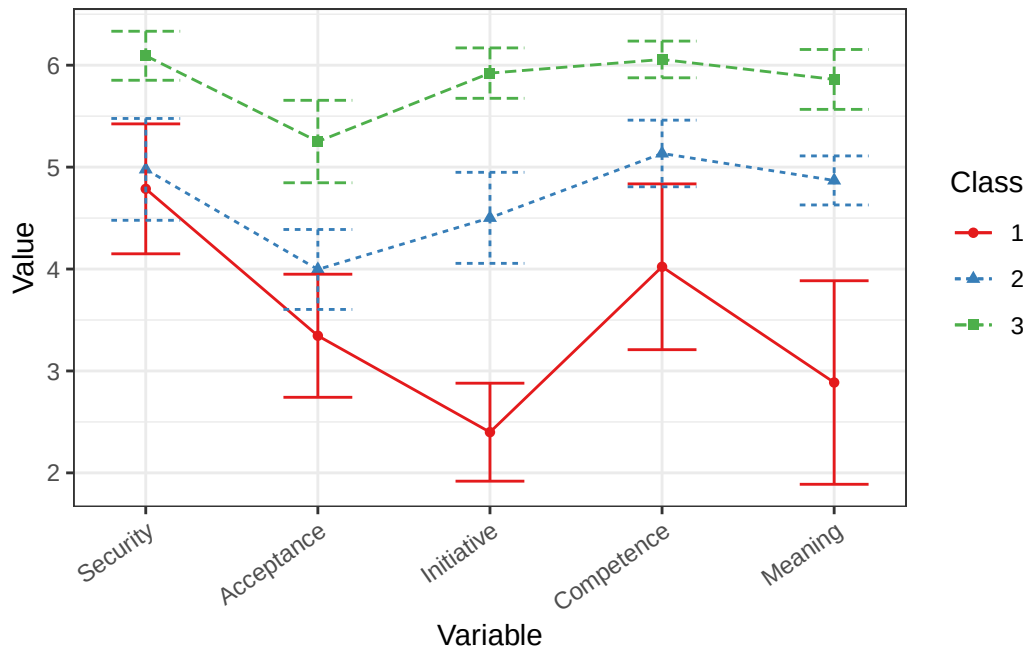
Warning: Using shapes for an ordinal variable is not advised



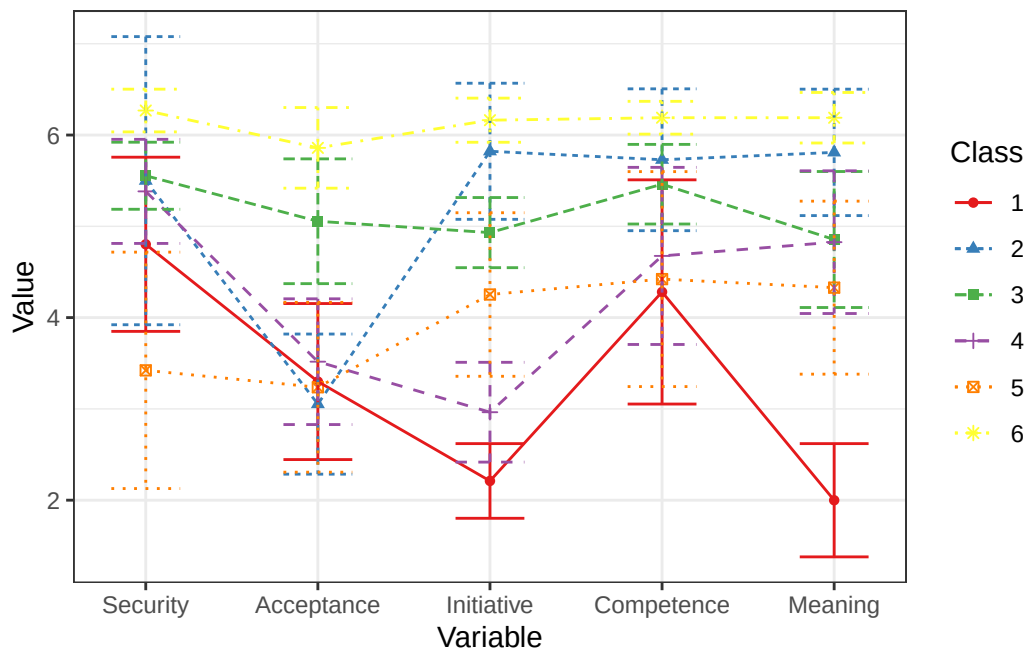
```
tidyLPA::plot_profiles(
  model1_3,
  rawdata = FALSE, # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE       # Hide standard-deviation bars.
) +
  ggplot2::theme(
    # Rotate and align indicator labels to improve readability.
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```



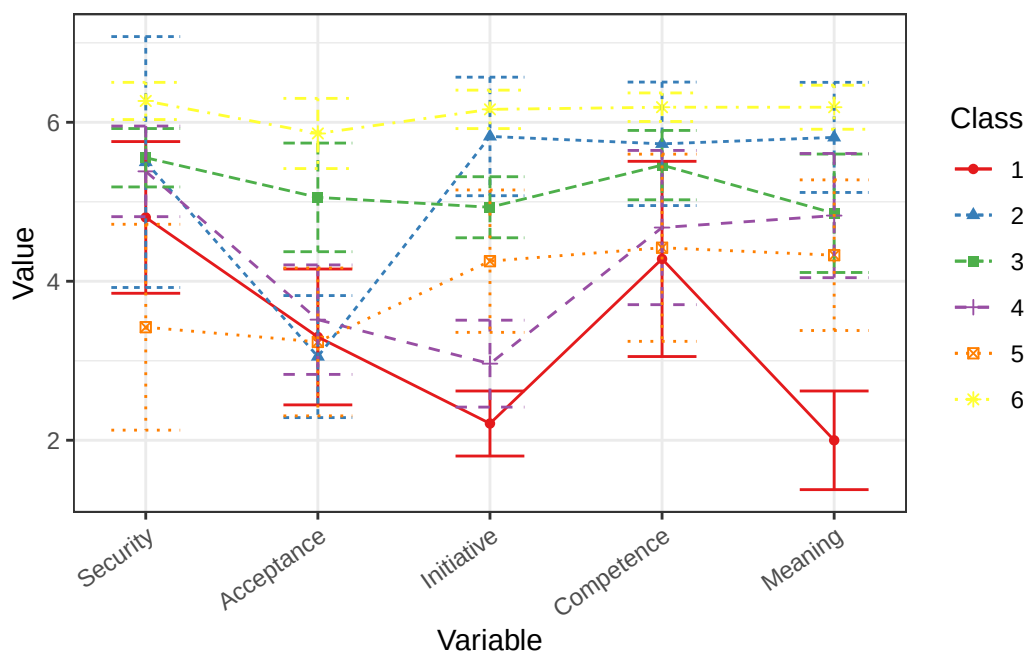
Warning: Using shapes for an ordinal variable is not advised



```
tidyLPA::plot_profiles(
  model1_6,
  rawdata = FALSE, # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE      # Hide standard-deviation bars.
) +
ggplot2::theme(
  # Rotate and align indicator labels to improve readability.
  axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
)
```



Warning: Using shapes for an ordinal variable is not advised



Comparing classes in Model 2

```
#Model 2 -- 2,3
#####
#Analysing if different classes (solutions) have significant differences
#####

m2c2 <- model2_all$model_2_class_2$model # Solution with 2 classes; model 2
m2c3 <- model2_all$model_2_class_3$model # Solution with 3 classes; model 2

compare_lmr <- function(smaller, larger) {
  tidyLPA::calc_lrt(
    n = smaller$n,
    null_ll = smaller$loglik,
    null_param = smaller$df,
    null_classes = smaller$G,
    alt_ll = larger$loglik,
    alt_param = larger$df,

    alt_classes = larger$G
  )
}

compare_lmr(m2c2, m2c3)
```

Lo-Mendell-Rubin ad-hoc adjusted likelihood ratio test:

LR = 129.714, LMR LR (df = 11) = 122.850, $p < 0.001$

```
data$class2 <- as.factor( model2_all$model_2_class_2$model$classification)

data$class3 <- as.factor( model2_all$model_2_class_3$model$classification)

# Count participants assigned to each class in each solution.
table(data$class2)
```

```
1 2
168 222
```

```
table(data$class3)
```

```
  1    2    3  
127 201  62
```

Comparing Variables Across the Groups

```
#####  
#Do the following analysis on the class  
#####  
  
data$latent_groups <- as.factor(  
  model2_3$model_2_class_3$model$classification)  
  
table(data$latent_groups)
```

```
  1    2    3  
127 201  62
```

```
#####  
# comparing the variables in the final LPA across the groups  
#####  
  
data %>%  
  dplyr::select(  
    Security, Acceptance, Initiative, Competence, Meaning,  
    latent_groups) %>%  
  gtsummary::tbl_summary(  
  
    by = latent_groups,  
  
    type = list(  
      c(Security, Acceptance, Initiative, Competence, Meaning) ~ "continuous"),  
  
    statistic = gtsummary::all_continuous() ~ "{mean} ({sd})",  
  
    digits = gtsummary::all_continuous() ~ 2,
```

Comparison Across Three Classes	1 N = 127 ¹	2 N = 201 ¹	3 N = 62 ¹
Security	4.47 (1.48)	5.73 (0.88)	6.58 (0.56)
Acceptance	3.64 (1.39)	4.50 (1.39)	6.13 (0.71)
Initiative	3.53 (1.42)	5.28 (0.86)	6.31 (0.56)
Competence	4.48 (1.45)	5.68 (0.66)	6.44 (0.56)
Meaning	4.01 (1.62)	5.30 (0.90)	6.45 (0.53)

¹Mean (SD)

```
missing = "no"
) %>% gtsummary::modify_header(label = "**Comparison Across Three Classes**")
```

Prediction

```
lm1 <- lm(AR_2 ~ latent_groups, data = data)
anova(lm1)
```

Analysis of Variance Table

Response: AR_2

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
latent_groups	2	188.90	94.452	87.36	< 2.2e-16 ***
Residuals	387	418.42	1.081		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
emmeans::emmeans(lm1, pairwise ~ latent_groups,
  adjust = "tukey")
```

\$emmeans

latent_groups	emmean	SE	df	lower.CL	upper.CL
1	4.33	0.0923	387	4.15	4.51
2	5.54	0.0733	387	5.39	5.68
3	6.27	0.1320	387	6.01	6.53

Confidence level used: 0.95

```
$contrasts
  contrast          estimate    SE  df t.ratio p.value
latent_groups1 - latent_groups2  -1.207 0.118 387 -10.237 <0.0001
latent_groups1 - latent_groups3  -1.943 0.161 387 -12.064 <0.0001
latent_groups2 - latent_groups3   -0.737 0.151 387  -4.878 <0.0001
```

P value adjustment: tukey method for comparing a family of 3 estimates

```
emmeans::eff_size(emmeans::emmeans(lm1, ~ latent_groups),
  edf = df.residual(lm1), sigma = sigma(lm1)
)
```

```
contrast          effect.size    SE  df lower.CL upper.CL
latent_groups1 - latent_groups2   -1.160 0.121 387   -1.398   -0.923
latent_groups1 - latent_groups3   -1.869 0.169 387   -2.201   -1.537
latent_groups2 - latent_groups3    -0.709 0.147 387   -0.999   -0.419
```

sigma used for effect sizes: 1.04
Confidence level used: 0.95

```
lm2 <- lm(AR_3 ~ latent_groups, data = data)
anova(lm2)
```

Analysis of Variance Table

Response: AR_3

```
      Df Sum Sq Mean Sq F value    Pr(>F)
latent_groups  2 102.51  51.255  36.246 4.518e-15 ***
Residuals    361 510.49   1.414
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
emmeans::emmeans(lm2, pairwise ~ latent_groups,
  adjust = "tukey")
```

```
$emmeans
latent_groups emmean    SE  df lower.CL upper.CL
1             4.36 0.1140 361    4.14    4.59
```


2	5.05	0.0849	361	4.88	5.21
3	5.98	0.1540	361	5.68	6.29

Confidence level used: 0.95

\$contrasts

contrast	estimate	SE	df	t.ratio	p.value
latent_groups1 - latent_groups2	-0.685	0.143	361	-4.805	<0.0001
latent_groups1 - latent_groups3	-1.622	0.191	361	-8.472	<0.0001
latent_groups2 - latent_groups3	-0.937	0.175	361	-5.343	<0.0001

P value adjustment: tukey method for comparing a family of 3 estimates

```
emmeans::eff_size(
  emmeans::emmeans(lm2, ~ latent_groups),
  edf = df.residual(lm2),
  sigma = sigma(lm2)
)
```

contrast	effect.size	SE	df	lower.CL	upper.CL
latent_groups1 - latent_groups2	-0.576	0.122	361	-0.815	-0.336
latent_groups1 - latent_groups3	-1.364	0.169	361	-1.696	-1.032
latent_groups2 - latent_groups3	-0.788	0.150	361	-1.084	-0.492

sigma used for effect sizes: 1.189

Confidence level used: 0.95